Jesus Andres Ruiz

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Education

University of Texas at San Antonio San Antonio, TX
Bachelor of Science in Mechanical Engineering | ***GPA: 3.6*** *August 2024 – May 2028*
UTSA Honors College

Relevant Experience

Intelligent Systems and Bionics Engineering Lab San Antonio, TX
Undergraduate Research Assistant *May 2026 – Present*

- Led fabrication of the OpenExo hip module, an open-source wearable exoskeleton, working closely with graduate students to fabricate and assemble hardware across 3D printing and machining.
- Supported a graduate student's Bowden cable transmission project, contributing to prototype design and assembly for ankle actuation.
- Diagnosed and resolved 3D printer hardware/adhesion issues across multiple printer platforms to maintain consistent prototyping output.

UTSA Hypersonics Lab San Antonio, TX
Undergraduate Research Volunteer *January 2026 – May 2026*

- Fabricated precision diaphragms in the machine shop for use in a Mach 7.2 Ludwieg Tube wind tunnel, supporting high-speed flow experiments and ensuring reliable burst pressures for test runs.
- Assisted researchers during hypersonic wind tunnel testing, helping prepare instrumentation, monitor runs, and collect experimental data from the wind tunnel.
- Set up and aligned laser-based Schlieren imaging systems to visualize shock waves and boundary-layer structures in hypersonic flow experiments.

UTSA Formula Society of Automotive Engineers San Antonio, TX
Mechanical Engineer / Team Member *August 2025 – Present*

- Designed and iterated mechanical components using SolidWorks, applying design for manufacturing principles and engineering drawings.
- Collaborated with multidisciplinary teams to refine designs under competition constraints.
- Translated theoretical engineering concepts into real-world vehicle systems.

Project Experience

Open Exo Hip Module San Antonio, TX
Undergraduate Research Assistant – Project Lead *May 2026 – Present*

- Re-Designed and iterated hip module components in Fusion 360, using FEA to validate redesigns after early prototype failures.
- Manufactured parts via 3D printing (FDM, continuous carbon fiber), CNC machining, and outsourced cutting, resolving tolerance/material issues.
- Iterated through multiple structural failure analyses (cracked cuffs, sliders, and carbon fiber struts) to redesign components for improved durability, incorporating targeted reinforcement in high stress zones.

Bowden Cable Transmission – Ankle Actuation San Antonio, TX
Undergraduate Research Assistant *May 2026 – Present*

- Prototyped a Bowden cable transmission enabling remote ankle actuation, reducing distal leg mass and metabolic cost of walking.
- Printed, and tested tapered vs. straight TPU cable inserts to optimize routing and fit for cables.

Skills

CAD: SolidWorks, Fusion 360
Programming / Embedded: Arduino, MATLAB, Python, Git/Github, ROS (learning)
Engineering: Technical documentation, design iteration, data evaluation, Microsoft office, manufacturing, FEA